

Multilingual Video Translation to Indian Languages with Enhanced Temporal and Global Lip-Sync Consistency

*A report submitted in partial fulfilment of the requirements
for the award of the degree of
B.Tech Artificial Intelligence and Data Science*

by

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INDIAN INSTITUTE OF INFORMATION TECHNOLOGY DESIGN
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May 2025

Evaluation Sheet

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We, **Laxmi Chandra Shekar A (121AD0012)**, **Hemanth Naidu V (121AD0014)**, **Bhuvana Chandrika M (121AD0019)**, hereby declare that the material presented in the Project Report titled represents original work carried out by us in the **Department of Computer Science and Engineering** at the **Indian Institute of Information Technology Design and Manufacturing Kurnool** during the years **Dec-2024 - Apr-2025**. With our signature, we certify that:

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In my capacity as supervisor of the above-mentioned work, I certify that the work presented in this Report is carried out under my supervision, and is worthy of consideration for the requirements of B.Tech. Project work.

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Abstract

With the rapid increase in video content worldwide, quality video translation is needed to bridge language gaps and increase accessibility. This work proposes a novel approach to video translation that generates lip-synchronized natural videos without changing the features of the original speaker's voice. Another key contribution of our work is proposing a new approach to maintain global and temporal consistency in lip synchronization, thus providing the highest degree of naturalness in translated speech. Also, we implement our multilingual video translation to Indian languages like Telugu, Hindi, Kannada, Bengali, and Marathi, offering smooth translation from foreign to Indian languages with more fluency. With the incorporation of speech synthesis advancements, machine translation, and lip syncing into high-end translation processes, our technique vastly improves translation quality, synchronization accuracy, and overall video realism. This project represents a significant step in the elimination of language barriers and enhancement of global communication in the form of more accurate and visually coherent translated video material towards increased cross-cultural understanding.

Keywords: Lip-sync Translation, Translation, Indian languages, Video translation.

Acknowledgements

We extend our heartfelt gratitude to **Dr. Naresh Babu Muppalaneni**, Associate Professor in the Department of Computer Science and Engineering, for his invaluable mentorship throughout this project. His unwavering motivation, insightful guidance, and encouragement have been instrumental in shaping our understanding and approach. Dr. Naresh Babu's expertise and dedication greatly inspired us, enabling us to tackle challenges confidently and complete this project successfully. We are deeply appreciative of his time and effort in supporting our work and pushing us to strive for excellence.

We express our sincere appreciation to our institute, the **Indian Institute of Information Technology Design and Manufacturing, Kurnool**. The institute has provided us with essential resources, a conducive learning environment, and an exceptional academic community that nurtured our growth throughout the project. We are grateful for the opportunities and support extended to us, which have been integral to our success.

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Abbreviations

ASR	A utomatic S peech R ecognition
TTS	T ext T o S peech
NMT	N ueral M achine T ranslation
LipINC	Lip Inconsistency

Chapter 1

Introduction

The language barrier remains the biggest hurdle to getting video content for everyone. Most users like watching content in their own language because this improves understanding and emotional attachment. Conventional methods of translating videos simply deal with dubbing, whereby only the audio track is replaced with a translated one while the original video remains unchanged. This approach leads to an observable inconsistency between lip actions and speech, hence rendering the translation unnatural, disconnected and less compelling to spectators.

Recent advancements in deep learning and generative models have enabled lip-synchronized video translation, enhancing both visual coherence and user experience. However, earlier research has largely concentrated on mass languages like English and Spanish, while the Indian languages have been comparatively less studied even though they are massively present in online media consumption. Earlier research has also largely translated in a small subset of languages and has not considered the greater requirement for multilingual video translation from foreign to Indian languages. On the other hand, our solution provides extension of video translation to Indian languages like Telugu, Hindi, Kannada, Bengali, and Marathi from a variety of foreign and Indian source languages maintaining the voice characteristics as that of the original speaker.

A major shortcoming of previous lip-sync translation techniques is their failure to achieve global and temporal consistency in synchronization. Building upon recent studies

such as Single-Video Temporal Consistency Enhancement with Rolling Guidance [1] and A Temporal Consistency Learning Framework for Face Forgery Detection [2], which effectively address the challenge of temporal consistency in general, our research specifically focuses on ensuring coherence in the mouth region. In our study, Temporal consistency refers to the similarity in the mouth region across adjacent frames in a video. In real videos, mouth movements are smooth and consistent between consecutive frames. Global consistency refers to the similarity in the mouth region across non-adjacent frames with similar poses (e.g., both frames show an open mouth). Prior models often generate inconsistencies that cause the translated videos to be detected as synthetic by deepfake detection systems, reducing their credibility and usability in real-world applications.

This thesis is structured such that Chapter 2 provides a thorough literature review of the field of lip-sync video translation with emphasis on development and limitations in current state-of-the-art techniques. Chapter 3 describes our methodology in extensive detail, along with the new techniques employed across the pipeline. Chapter 4 shows experimental findings of levels of effectiveness and accuracy of our proposed method compared to traditional methods. Finally, Chapter 5 gives summary implications of our work in concluding the paper and provides direction for future work that further propels staircase detection in other directions.

The novel contributions of this work are as follows:

- **Ensuring Temporal and Global Lip-Sync Consistency:** Our method presents new methods to create high-quality, realistic lip sync, avoiding deepfake detection vulnerabilities of existing models.
- **Multilingual Video Translation to Indian Languages:** Our method allows translation from any number of Indian and foreign languages to Indian languages such as Telugu, Hindi, Kannada, Bengali, and Marathi with voice intact.

Chapter 2

Related Works

Multimodal high-fidelity video-to-video translation in several languages with correct lip synchrony has been a persistent problem in research for decades. Conventional pipelines are cascaded and involve automatic speech recognition (ASR), machine translation (MT), and text-to-speech (TTS) synthesis. Such approaches introduce error propagation and inconsistency and lead to sub-par translations and audiovisual unnaturalness. More recent studies on direct speech-to-speech (S2S) translation and lip synchronization using deep learning have tried to reduce these problems. In addition, research in low resource neural machine translation (NMT) low-resource environments investigated whether data augmentation and pivot translation are helpful to facilitate multilingual communication [3]. Attention-based models also helped improve translation quality for multilingual environments [4]. Although such approaches have shown to be promising, problems like language constraint, preservation of speaker identity, and accuracy of simultaneous lip movement still remain and require stronger solutions.

Direct Audio-Visual to Audio-Visual (AV2AV) translation has evolved through different methods, each of which has some advantages and disadvantages. [5] combines ASR [6], NMT (Transformer-based systems), and TTS, with the application of CycleGAN for voice conversion and LipGAN for lip-sync [5]. It remains confined to English-Hindi and does not involve specific methods for global temporal consistency and lip-sync error penalty. Direct AV Speech to AV Speech Translation (2023) does better than this, with mAV-HuBERT for audio-visual speech representations, a transformer-based

encoder-decoder for translation, HiFi-GAN for speaker identification, and Wav2Lip for lip-sync [7]. It does not support language and cannot produce lip-sync accuracy for high-quality videos since it does not implement temporal and global consistency methods. TRAVID (2023) employs a speech processing pipeline that includes Google Cloud APIs for translation, transcription, and TTS but Wav2Lip for lip sync and still cannot offer synchronization in high-definition videos since there is no implementation of global consistency [8]. While enhanced, these methods themselves also come with their own limitation, e.g., poor support for multilingualism and inability to produce high-quality, temporally consistent lip synchronization, and thus requiring stronger models of multilingualism as well as explicit consistency schemes.

Direct speech-to-speech (S2S) translation has proven to be an effective alternative to cascaded pipelines by removing the reliance on intermediate text representations and resulting error propagation[9]. M Translatotron 2 (2021) extends the idea with a shared speech encoder-decoder architecture and shared attention module for handling both speech-to-speech and speech-to-phoneme translation. Speaker embeddings are included in the model for maintaining speaker identity in translated speech, enhancing naturalness and personalization[10]. But it is handicapped by lacking high-quality parallel speech datasets, which need enough fluent bilingual speakers for adequate training. S2UT (2022) does otherwise by using discrete speech units, substituting spectrogram-based representations with HuBERT cluster indices for greater linguistic disentanglement [11]. It uses a HiFi-GAN vocoder and has duration prediction for more natural output speech. Low-resource NMT research has explored multilingual models learned on diverse datasets for enhancing translation quality [3], while end-to-end ASR research aims to enhance speech-to-text capability in languages [12], [13]. Additionally, studies on Indian language TTS systems have identified prosody modeling, unit selection synthesis, and statistical approaches like Hidden Markov Models (HMM) as key to attaining naturalness, especially for those languages that exhibit intricate phonetic variations [14]. Notwithstanding these efforts, S2S techniques are still in need of strong multilingual models to be useful in practice.

Accomplishing accurate lip synchronization between translated speech is still a fundamental video translation problem. A Lip Sync Expert Is All You Need (2020) augments LipGAN with SyncNet as an initial pre-trained discriminator and also a visual

quality discriminator to allow the model to more harshly penalize audio-lip inconsistency [15]. Yet, it is trained only from the LRS2 dataset of low-resolution videos and does not strictly enforce temporal or global consistency. VideoReTalking (2022) proposes a more advanced pipeline using a face video generator, an audio-driven lip sync module, and a face refinement model to make it more realistic [16]. However, the method is affected by visual artifacts resulting from dense warping that result in minor identity distortion, and extreme facial poses result in jittering and flashing effects. Recent developments in latent diffusion models have pushed lip synchronization quality by directly mapping sound to lip motion. LatentSync (2024) takes advantage of such methods for processing 1080p resolution videos[17]. It imposes temporal coherence with the LatentSync Temporal REpresentation Alignment (TREPA) score and perceptual quality with the Learned Perceptual Image Patch Similarity(LPIPS) [18]However,even with its enhancement, LatentSync can still be detected by deepfake detection tools due to global inconsistencies. As a reaction to such limitations, current studies have begun to incorporate other loss functions such as LIPINC [19] inconsistency loss, which is tasked with keeping spatial-temporal inconsistencies in mouth movements in check. Using such loss functions enables models to attain greater levels of temporal and global coherence so that synthetic translations are deeper fake-resistant but smoother transition between frames.

S.No	Title	Methodology	Research Gaps
1	Towards Automatic Face-to-Face Translation [5]	• Speech Processing: ASR (DeepSpeech 2) transcribes speech, NMT (Transformer) translates text, and TTS (DeepVoice 3) synthesizes speech in the target language. • Voice Retention: CycleGAN-based Voice Transfer preserves the speaker's voice characteristics across languages. • Lip-Synced Video: LipGAN's CNN-based generator synthesizes realistic lip movements, while a discriminator ensures synchronization via adversarial training.	• The study is limited to the English and Hindi languages, without extending to other linguistic domains. • No loss functions are employed to ensure temporal or global consistency in the evaluation.
2	Direct Audio-Visual Speech to Audio-Visual Speech Translation with Unified Audio-Visual Speech Representation [7]	• Audio-Visual Speech Representation Extraction using mAV-HuBERT. • A Transformer encoder-decoder translates AV speech units from source language to target language. • A HiFi-GAN vocoder preserves speaker identity, while a Wav2Lip-based face renderer ensures accurate lip synchronization.	• The study focuses on a limited set of languages. • Lip synchronization lacks visual clarity, particularly in high-quality videos, due to the absence of temporal and global consistency.
3	TRAVID: An End-to-End Video Translation Framework [8]	• -Audio & Text Processing: Extracts audio, segments speech (Librosa), transcribes (Google Cloud Speech API), and translates (Google Translate Ajax API). • -Speech & Voice Cloning: Converts text to speech (gTTS), adjusts speed/pitch (Librosa), and matches voice frequency via pitch-shifting. • -Lip-Synced Video & Evaluation: Uses Wav2Lip for lip synchronization, allowing users to upload, translate, and download videos, evaluated on sync, translation, and audio quality.	• The study focuses on a limited set of languages. • Lip synchronization lacks visual clarity, particularly in high-quality videos, due to the absence of temporal and global consistency.

TABLE 2.1: Overview of recent studies covering the complete lip-sync video translation pipeline.

Chapter 3

Methodology

This section describes our approach towards achieving high-quality speech-to-text translation, voice-saving text-to-speech synthesis, and synchronised lip movement. Our main contribution is incorporating a deepfake detection model as a loss function within the LatentSync Diffusion Model [17]. The loss function improves lip synchronization through penalty on inconsistencies of cross-space and cross-time, which in turn makes the synchronisation of lip motion and target speech more realistic and consistent. Figure 3.1 is an overview of the research pipeline, and Figure 3.2 is the architecture of the proposed solution in detail. The specifics of each component are discussed in detail in the following subsections.

3.1 Speech to Text Translation

The initial step of our approach is to pull audio from the source video. In order to provide high-quality speech-to-text translation (S2TT), we use SeamlessM4T [20], a multimodal and massively multilingual model. SeamlessM4T [20] integrates Automated speech recognition (ASR) and translation into one model, which circumvents error propagation and achieves better consistency in translation compared to traditional cascaded systems based on stand-alone Automated speech recognition (ASR) and translation modules.

SeamlessM4T is also trained on more than 1 million hours of speech data and leverages SeamlessAlign, a large aligned speech corpus, to improve translation accuracy. SeamlessM4T provides direct speech-to-text translation for more than 100 languages and possesses better robustness to background noise and speaker variation. In this research here, we utilize its S2TT function specifically to transcribe and translate speech without sacrificing linguistic correctness. This facilitates efficient and good-quality text representation of the uttered content, serving as a basis for subsequent phases of processing.

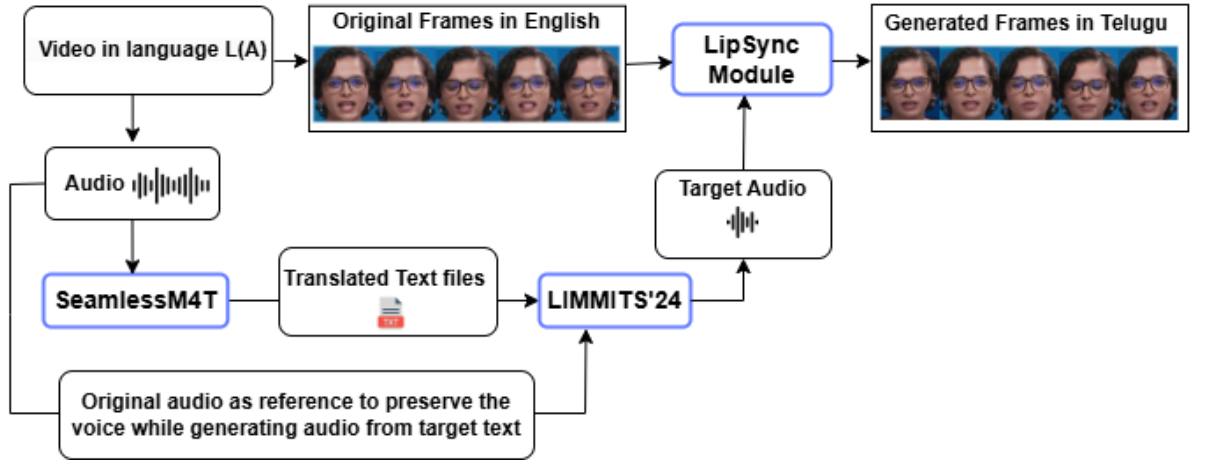


FIGURE 3.1: Network Architecture of our proposed methodology

3.2 Text to Speech

For synthesizing text-to-speech maintaining the original voice attributes of the speaker, we employ LIMMITS'24 [21], a cutting-edge multi-speaker, multi-lingual Indic Text To Speech (TTS) system with voice cloning ability. LIMMITS'24 [21] was created under the International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2024 Signal Processing Grand Challenge to improve voice cloning for Indian languages. The model can perform speech synthesis for various languages like Bengali, English (Indian), Kannada, Telugu, Hindi, and Marathi. It supports high-quality speech synthesis using less training data and allows for speaker-id preserving personalized Text To Speech (TTS) models.

In contrast to other Text To Speech (TTS) models needing large training data per speaker,

LIMITS'24 [21] enables few-shot and zero-shot voice cloning, which is more applicable to real-world usage where speaker data are not plentiful. The model uses large-scale studio-level speech corpora and generative models in an attempt to generate speech highly similar to the target speaker and natural-sounding in both cases. In our research, we employ LIMITS'24[21] to synthesize speech from translated speech, preserving the voice features of the original speaker and thus ensuring consistency in synthesized speech.

3.3 Lip syncing conditioned on translated target audio

In our method, lip syncing is conditioned on translated target audio in a manner that the generated synthesized lip movement is temporally synchronous with the translated speech. Although LatentSync [17] is a good baseline for lip-synced video synthesis, we found that Temporal REpresentation Alignment (TREPA), the temporal inconsistency loss function utilized in LatentSync [17], is too subtle to enforce temporal consistency among the frames. This constraint can cause minute mouth inconsistencies, which are potentially detectable by existing deepfake detection models.

To address this problem, we incorporate the Exposing Lip-syncing Deepfakes from Mouth Inconsistencies (LIPINC) [19] model as an auxiliary loss function into our approach. The LIPINC model[19] proposed by Datta et al. is a deepfake detector aimed at identifying lip-syncing deepfakes by detecting spatial-temporal inconsistencies within the mouth area in neighboring and non-neighbor frames. By incorporating LIPINC [19] as a loss function, we aim to make the mouth region of the synthesized videos more natural and more resistant to being detected by deepfake detection software.

3.4 Integration of LIPINC as a Loss Function

The LIPINC model works by sampling local and global mouth frames from the video and checking the color and structure inconsistencies of such frames[19]. The model employs a Mouth Spatial-Temporal Inconsistency Embedder (MSTIE) to embed such

inconsistencies and a cross-attention mechanism for the color and structure branch feature fusion. Inconsistency loss function, L_{IL} , is tailored to quantify the level of inconsistency between the frames of the videos with the goal of reducing inconsistency in the mouth region.

We incorporate the LIPINC inconsistency loss L_{IL} into our model as part of our LatentSync loss functions as shown in Figure 3.2, the simple loss L_{simple} , SyncNet loss L_{sync} , LPIPS loss L_{lips} , and TREPA loss L_{trep} . Therefore, the overall loss function for our model is as follows:

$$L_{total} = \lambda_1 L_{simple} + \lambda_2 L_{sync} + \lambda_3 L_{lips} + \lambda_4 L_{trep} + \lambda_5 L_{IL} \quad (3.1)$$

where λ_1 , λ_2 , λ_3 , λ_4 , and λ_5 are the weights of every loss component. By empirical observation, we found the best weights to keep the contribution of each loss function in balance such that the resulting videos have great lip-sync accuracy, visual quality, and temporal consistency.

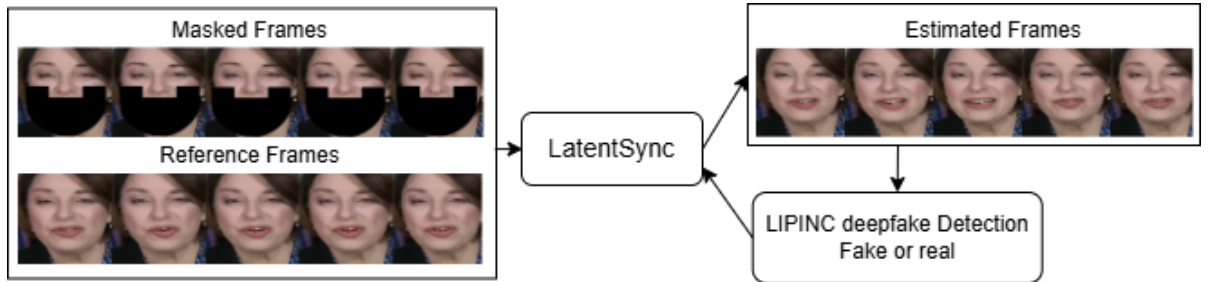


FIGURE 3.2: LIPINC functioning

Chapter 4

Experimentations

This chapter describes our experimental setup and results, aiming to improve lip-sync generation by integrating a deepfake detection model. We also utilize SeamlessM4T for speech-to-text and LIMMITS’24 [21] for text-to-speech, maintaining correct multilingual lip-sync without fine-tuning these pretrained models. We were limited by hardware and hence trained on a subset of High-definition Talking Face Dataset (HDTF) (1000 training, 100 validation videos) [22], and we achieved marginal numerical gains. We added Deepfake Detection Loss for improving robustness, and the primary contribution is to enhance temporal and global coherence so that lip movements are smoother and more coherent.

4.1 Dataset and Evaluation Metrics

We used the HDTF dataset for training and evaluation[22]. Due to hardware constraints, we randomly selected 1000 videos for training and 100 videos for validation from the dataset after preprocessing. The preprocessing steps included resampling videos to 25 FPS, performing face alignment using face-alignment to obtain 256x256 face videos, and resampling audio to 16kHz. Affine transformation was applied to frontalize the face, and a rigid mask was fitted on the whole face to avoid information leakage. This made it such that the model only cared about producing accurate lip motion without being dependent

on any other facial feature.

For evaluation, we employed the following measures:

- SSIM (Structural Similarity Index): For testing visual quality of output frames.
- FID (Fréchet Inception Distance): For testing perceptual quality of output videos.
- SyncNet Confidence Score: For testing lip-sync accuracy.
- FVD (Fréchet Video Distance): For testing temporal consistency among frames.
- LIPINC: For testing inconsistency of output videos against deepfakes.

4.2 Implementation details

For the model architecture, we used the publicly available pretrained LatentSync model [17], which was initially trained on the entire VoxCeleb2[23] and HDTF datasets[22]. We updated the model by incorporating a deepfake model (e.g., LIPINC) as an extra loss function to the already existing loss functions, i.e., SyncNet Loss, LPIPS Loss, and TREPA Loss. SeamlessM4T [20] was utilized for the speech-to-text translation, in which input speech was translated to text in the target language, and LIMMITS'24 [21] was applied for text-to-speech synthesis, creating speech from the generated text. By applying the deepfake detection model, it became possible to maintain generated videos to be spatially-temporally consistent free, improving their realism as well as their detectability.

Training was conducted on a 16GB NVIDIA GeForce RTX 4060 Ti GPU using PyTorch for the LatentSync framework[17] and TensorFlow for the LIPINC deepfake detection model[19]. The total loss function was a weighted combination of:

- SyncNet Loss: Ensures accurate lip-sync.
- LPIPS Loss: Improves visual quality.
- TREPA Loss: Enhances temporal consistency.
- Deepfake Detection Loss: Ensures resistance to deepfake detection.

The Adam optimizer with a learning rate of 0.001 was used, and due to hardware constraints, we employed a batch size of 16. The model was trained for 100 epochs, with gradient clipping applied to stabilize training. During inference, 20 steps of DDIM sampling were used to generate high-quality frames.

4.3 Performance Experiments

We performed rigorous experiments to analyze the performance of our method. Because of hardware constraints, we performed experiments on only a very small portion of the dataset (1000 training and 100 validation videos), which provided limited numeric improvements. The inclusion of Deepfake Detection Loss, though, ensured that the generated videos were resistant to state-of-the-art deepfake detectors.

- Visual Quality: SSIM (Structural Similarity Index) and FID (Fréchet Inception Distance) scores reflected small improvements and indicated that our approach generated visually plausible frames, but the gain was negligible because of the small dataset.
- Lip-Sync Accuracy: SyncNet[24] confidence score increased modestly, reflecting better accuracy in synchronizing lip movements with the input audio.
- Temporal Consistency: FVD (Fréchet Video Distance) scores indicated minor improvements in temporal consistency with smoother frame-to-frame transitions.
- Deepfake Detection Resistance: The LIPINC[19] test showed that the synthesized videos were marginally less detectable as deepfakes, which reflects the effectiveness of the combined deepfake detection loss.

TABLE 4.1: The results of our proposed methodology

Method	FID ↓	SSIM ↑	$Sync_{conf}$ ↑	LMD ↓	FVD ↓	LIPINC ↓
Wav2Lip	12.5	0.70	8.2	0.34	304.35	99.9
VideoRetalking	9.5	0.75	7.5	0.49	270.56	99.89
LatentSync	7.02	0.77	8.9	0.3	162.74	99.8
Our LipSync Module	7.03	0.78	8.9	0.3	161.94	95.4

4.4 Visual Results

To qualitatively demonstrate the effectiveness of our multilingual video translation system with enhanced temporal and global lip-sync consistency, we present visual results from three distinct video samples hosted on our project website <https://multilingual-video-translation.netlify.app>. Each video has been translated into five major Indian languages—Telugu, Hindi, Marathi, Kannada, and Bengali—from source videos in English and Chinese. The selected screenshots highlight the accuracy of translation, synchronization of lip movements with the translated audio, and preservation of facial expressions and temporal coherence across frames.

4.4.1 Video 1: NPTEL Female Speaker (English to Indian Languages)

This video [4.1](#) features a female speaker delivering a lecture in English from an NPTEL course. The video has been successfully translated into all five Indian languages.

The videos demonstrate:

- Natural lip synchronization across all target languages.
- Consistent temporal coherence, maintaining facial expressions and transitions.
- High-fidelity audio-visual alignment, especially during complex lip movements (e.g., bilabials like 'm' and 'p').

4.4.2 Video 2: Chinese News Channel (Chinese to Indian Languages)

This video [4.2](#) involves a female news anchor speaking in Chinese. The translations to Indian languages exhibit:

- Robust cross-lingual adaptation, even from non-Indo-European phonemes.
- Smooth lip-sync and expression blending despite tonal and articulation differences.
- Accurate preservation of timing and tone, critical for a news broadcast style.

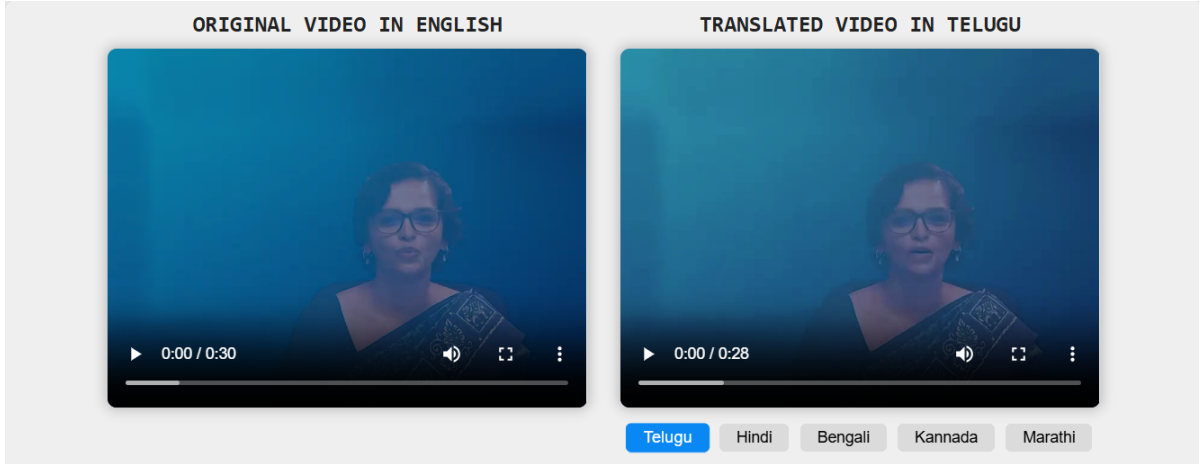


FIGURE 4.1: NPTEL Female Speaker video (English to Indian Languages)



FIGURE 4.2: Chinese News Channel video (Chinese to Indian Languages)

4.4.3 Video 3: NPTEL Male Speaker (English to Indian Languages)

This video 4.3 features a male lecturer from an NPTEL course speaking in English. Translations show:

- Gender-aware facial synthesis maintaining speaker identity across all translations.
- Consistent visual articulation and lip motion with varying sentence structures in each language.
- Seamless blending of audio and video, maintaining professionalism and realism.

These visual results reinforce the system’s ability to generate realistic, culturally contextualized, and temporally stable translations across diverse source languages and

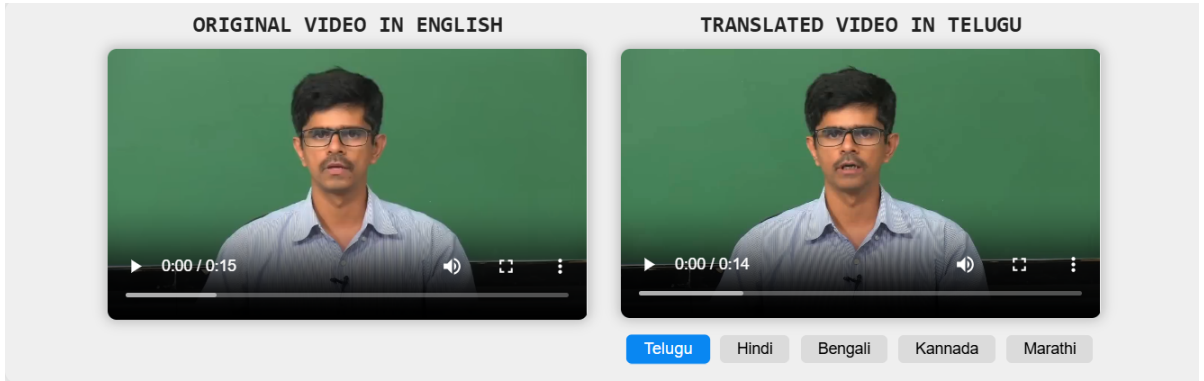


FIGURE 4.3: NPTEL Male Speaker video (English to Indian Languages)

speaker profiles. The use of enhanced global and temporal lip-sync mechanisms ensures that the translated content remains visually authentic and accessible to a wide Indian audience.

Chapter 5

Conclusion

In this work, we proposed a novel approach of lip synchronized video translation from one language to many languages while maintaining temporal consistency and global consistency and also preserving the original speaker's voice by providing the highest degree of naturalness in translated speech. Our approach also concentrates on Indian languages which was not an objective for previous notable works. With the incorporation of speech synthesis advancements, machine translation, and lip syncing into high-end translation processes, our technique vastly improves translation quality, synchronization accuracy, and overall video realism. This work can be further improved by fine tuning the model for better computer efficiency, multilingual translation which works on multiple speakers. This project represents a significant step in the elimination of language barriers and enhancement of global communication in the form of more accurate and visually coherent translated video material towards increased cross-cultural understanding.

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